When the Spleen Stops Building

Stromal remodelling and the loss of blood and
and immune cell production in spaceflight

Kireeti, Pranav, Aryan, Vedant

The Spleen Machine Interlake High School, Bellevue WA

Rodent Research 9

BACKGROUND

Astronauts come home immunosuppressed

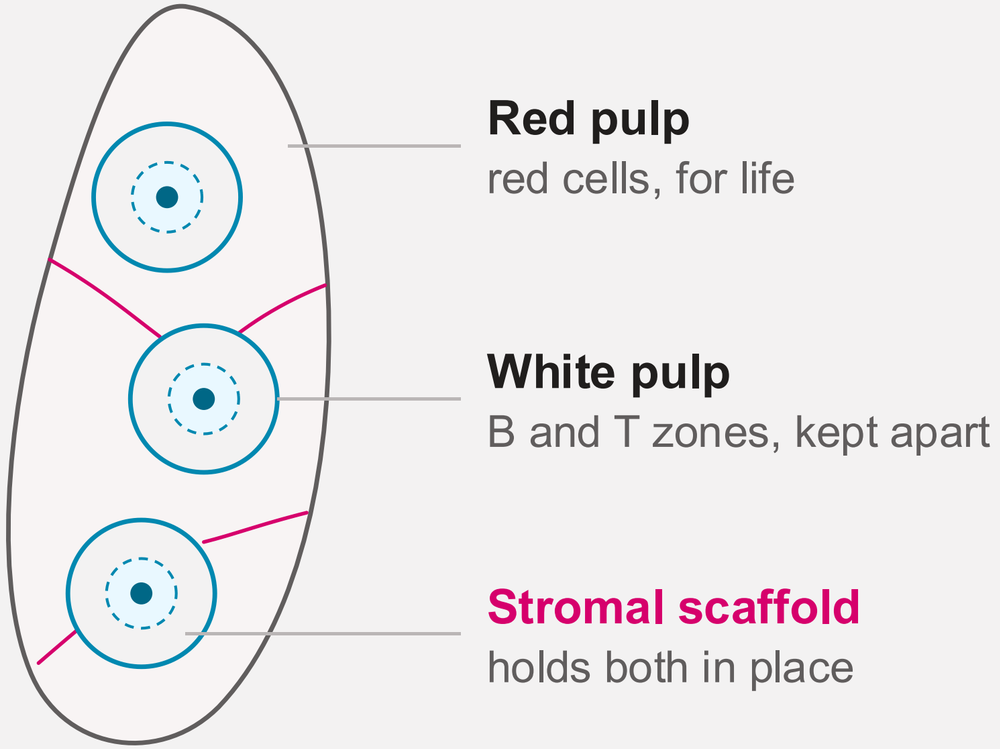
Spaceflight shrinks the mouse spleen

Every leukocyte population drops

Why is it unresolved?

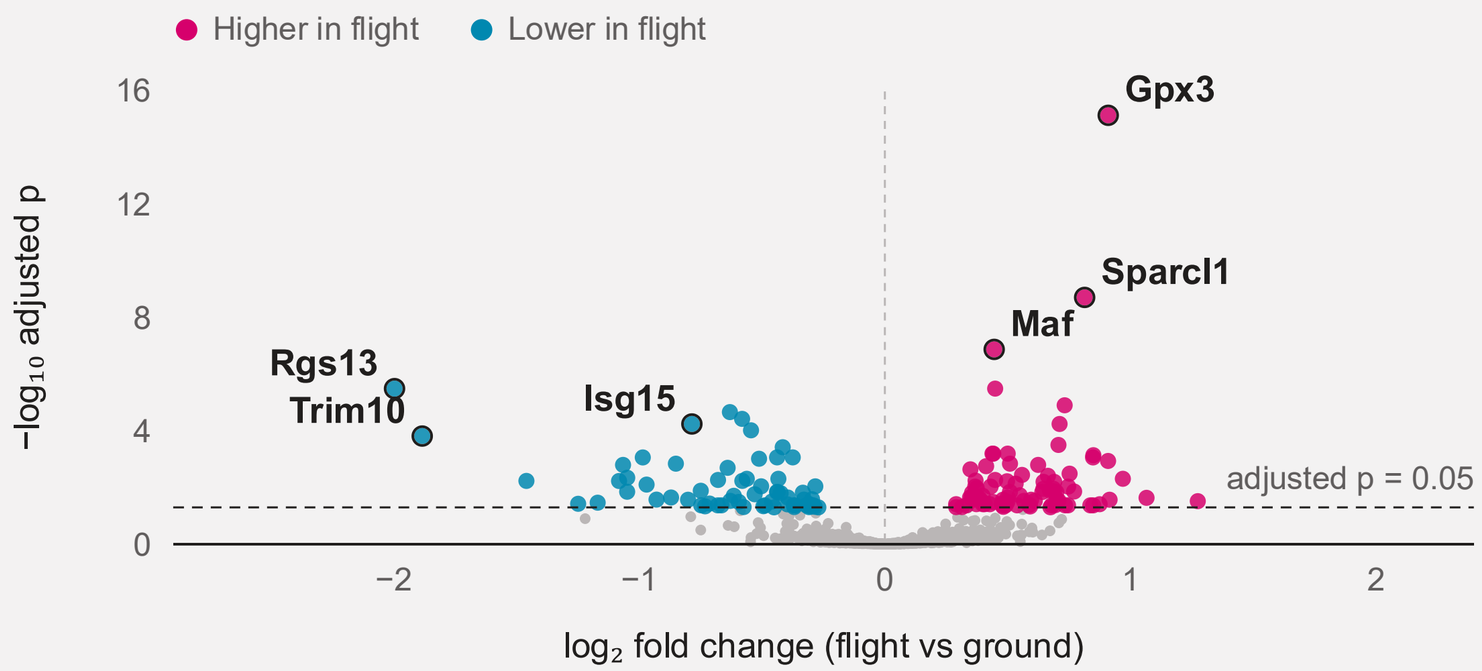
Are the immune cells damaged?
Or is the scaffold remodeled?

RR-9 spleen came through the Biospecimen Sharing Program and has never been analyzed on its own.



PRELIMINARY DATA

One gene stands out



146

genes at adjusted $p < 0.05$

10 v 10

flight vs ground control

33 d

on the ISS

Gpx3: strongest in the dataset at adjusted $p = 7 \times 10^{-16}$.

Figure 1. DESeq2 across all 14,475 tested genes; points above the dashed line pass FDR 0.05

PRELIMINARY DATA

Three modules move together

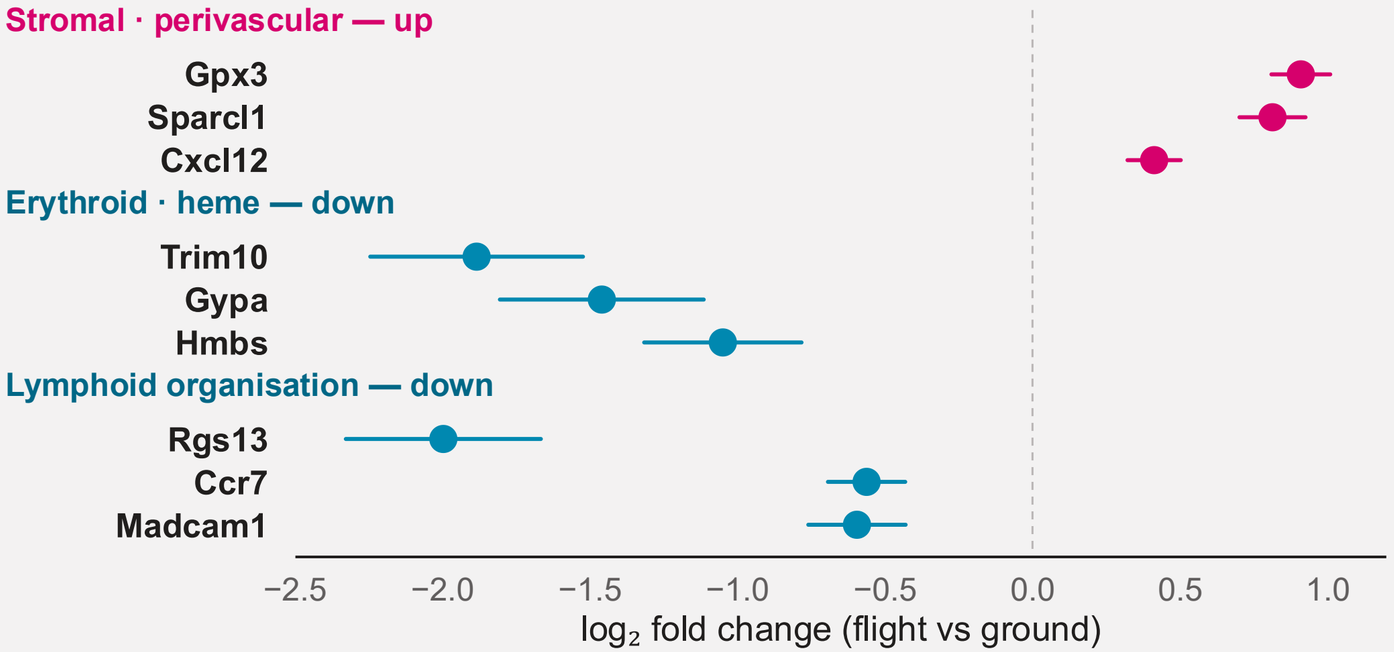


Figure 2. Three representative genes per module; bars are standard errors; hollow dots are not individually significant. Redrawn from NASA OSDR OSD-420.

Stromal · perivascular

up · $p = 1.3 \times 10^{-7}$

Gpx3 · Sparcl1 · Cxcl12

Erythroid · heme

down · $p = 2.3 \times 10^{-5}$

Trim10 · Gypa · Hmbs

Lymphoid organisation

down · $p = 0.0059$

Rgs13 · Ccr7 · Madcam1

The scaffold rises while both things it supports fall, not the pattern of an organ simply losing mass. Interferon tone also falls ($p = 6.6 \times 10^{-7}$), replicating Wu 2024.

HYPOTHESIS

“If spaceflight remodels the splenic stromal niche, then erythropoietic output falls, germinal-center organization is disrupted, and the stroma changes actively, not in proportion to lost lymphoid mass.”

AIM 1
Output

AIM 2
Organization

AIM 3
Mechanism

AIM 1 - OUTPUT

Lost erythropoiesis or altered composition?

Hindlimb unloading
28 d · both sexes

Perfuse
then weigh

Stage progenitors
three readouts

Expected
progenitors per gram

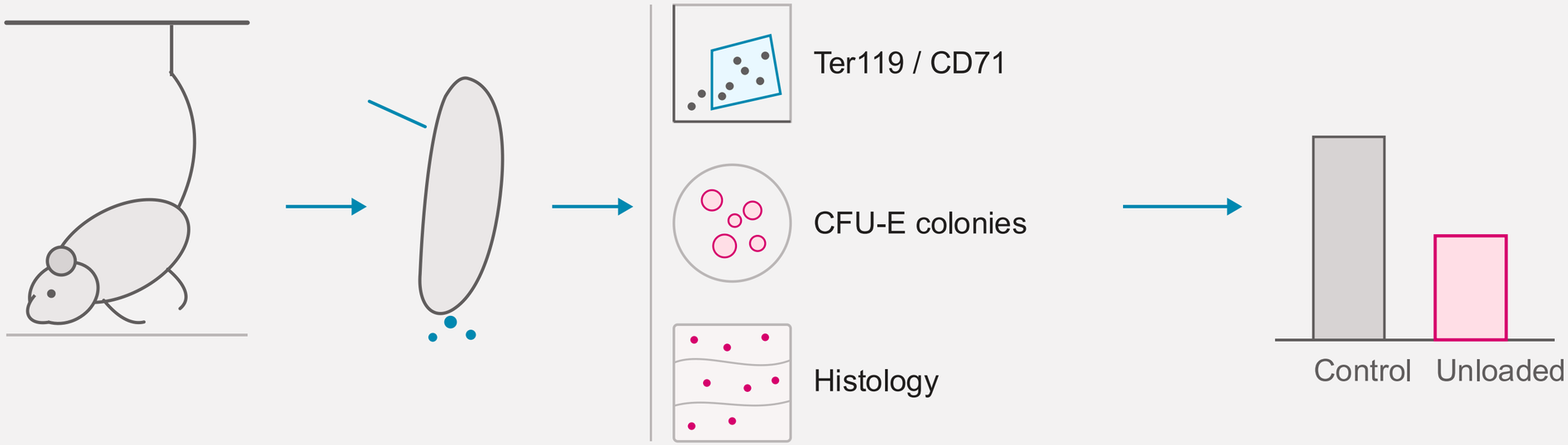


Figure 4. Aim 1 design. Progenitors are counted per gram of perfused tissue.

Bulk RNA-seq cannot separate lost production from altered composition. A per-gram count can.

AIM 2 - ORGANIZATION

Which cell makes Gpx3?

Locate Gpx3
snRNA-seq · RNAscope

Map the zones
immunofluorescence

Immunise
NP-KLH

Expected
germinal centres

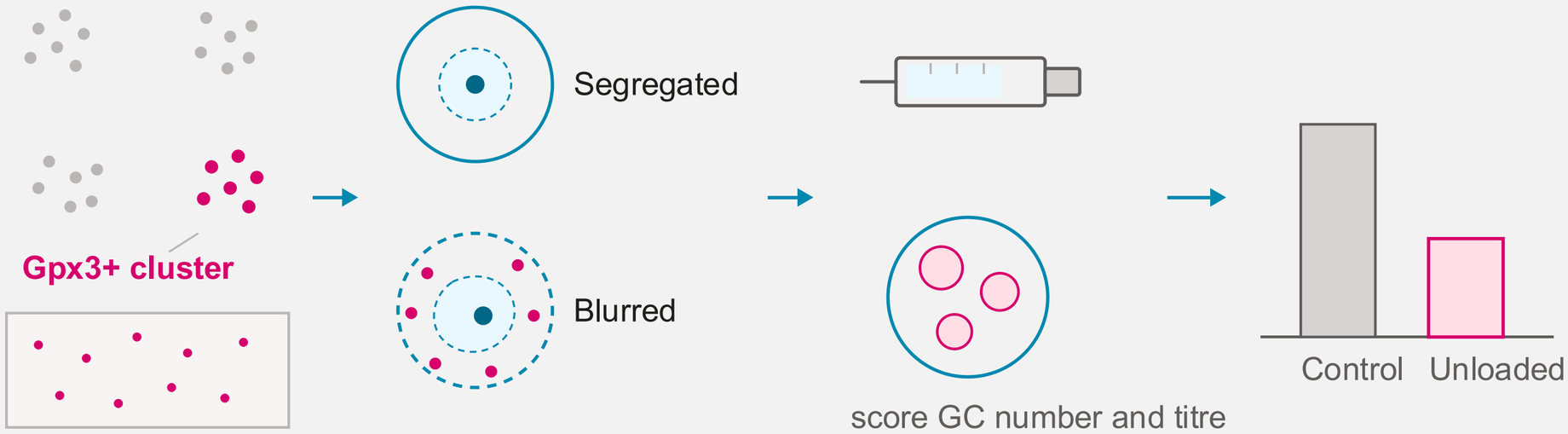


Figure 5. Aim 2 design. Locate the Gpx3-expressing cell, then test whether the zones still work.

AIM 3 - MECHANISM

Can selenium preserve output?

Two diets
adequate or supplemented

Unloading
28 d

Endpoints
three measures

Expected
progenitors retained

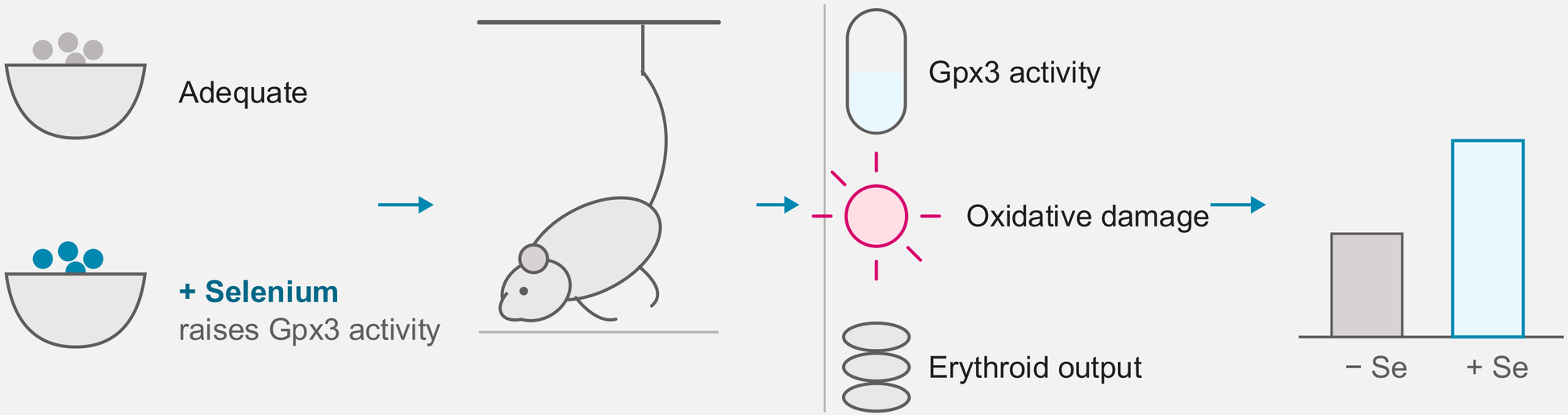


Figure 6. Aim 3 design. Two diets through unloading, read out three ways.

CONCLUSION AND SIGNIFICANCE

Where does immune decline begin?

The stroma changes actively. It does not simply shrink.

Gpx3 is new to spaceflight

Three aims separate cause from composition

The same processes run in immune ageing and chronic stress, so the answer does not stay in orbit.

THANK YOU

Acknowledgments

GL4HS and Ames Research Center

NASA OSDR — OSD-420, open data

RR-9 Biospecimen Sharing Program — the tissue

Questions?

1 Gridley et al. *PLoS One* 2013 — mouse thymus and spleen after STS-135

2 Pecaut et al. *PLoS One* 2017 — immune dysfunction and metabolism

3 Mebius & Kraal. *Nat Rev Immunol* 2005 — structure and function of the spleen

4 Inra et al. *Nature* 2015 — a perisinusoidal niche in the spleen

5 Wu et al. *Nat Commun* 2024 — conserved immune dysfunction in microgravity

6 Galazka et al. NASA OSDR 2021 — RR-9 spleen transcriptional profiling

Full eleven-reference list in the written Specific Aims document.